

ANSWER KEY

GATE CS — GATE CS 2026 Trend-Based Mock Paper
3

Q.No.	Type	Answer	Marks
1	3MCQ	A	3
2	2MCQ	C	2
3	3MCQ	B	3
4	2MCQ	C	2
5	2MCQ	24	2
6	2MCQ	A	2
7	2MCQ	B	2
8	2NAT	B	2
9	2MSQ	C	2
10	2MCQ	B	2
11	2MCQ	B	2
12	2NAT	B	2
13	2NAT	C	2
14	1MCQ	C	1
15	1MCQ	B	1
16	1MCQ	A	1
17	1MCQ	B	1
18	1MCQ	B	1
19	1NAT	C	1
20	1MCQ	B	1
21	1MCQ	C	1
22	2NAT	C	2
23	1NAT	C	1
24	1MCQ	B	1
25	1NAT	A	1
26	2MCQ	C	2
27	1MCQ	A	1
28	1MCQ	C	1
29	1MCQ	B	1
30	1MCQ	B	1
31	1MCQ	B	1
32	1NAT	B	1
33	1MCQ	C	1
34	1NAT	C	1
35	1NAT	B	1
36	1MCQ	B	1

37	1MCQ	B	1
38	1NAT	A	1
39	1MCQ	B	1
40	1MCQ	B	1
41	1NAT	B	1
42	1MCQ	B	1
43	1MCQ	B	1
44	1MCQ	C	1
45	1MCQ	A	1
46	2NAT	12.20.164.0/22, 12.20.168.0/22	
47	2MCQ	B	2
48	2MSQ	B	2
49	2NAT	System call, hardware interrupt, exception (trap)	2
50	2MCQ	D	2
51	2MCQ	B	2
52	2MCQ	B	2
53	2MCQ	A	2
54	2MCQ	B	2
55	2MSQ	A	2
56	1MCQ	B	1
57	2MCQ	B	2
58	2NAT	T1 !' T3 !' T4 !' T2	2
59	2MCQ	A	2
60	2NAT	1	2
61	2MCQ	3	2
62	2NAT	11	2
63	1MCQ	A	1
64	2MSQ	D	2
65	2MSQ	B	2

Q1. General Aptitude | 3 marks | 3MCQ

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is re...

Answer: A

From map: Library is northwest of Canteen. This matches option A.

Source: Adapted from GATE 2025 CS Session 1 Q5

Q2. General Aptitude | 2 marks | 2MCQ

If '' denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered _...

Answer: C

Walk < jog < sprint (increasing intensity). Bothered < fazed < daunted. 'Fazed' means disturbed, matching the intensity progression.

Source: Adapted from GATE 2024 CS Session 2 Q1

Q3. General Aptitude | 3 marks | 3MCQ

In the sequence 6, 9, 14, x, 30, 41, a possible value of is

Answer: B

Differences: 9-6=3, 14-9=5, x-14=?, 30-x=?, 41-30=11. Pattern: 3,5,7,9,11. So x-14=7, giving x=21.

Source: Adapted from GATE 2024 CS Session 2 Q5

Q4. General Aptitude | 2 marks | 2MCQ

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of S...

Answer: C

Total 10000. Neither = 1500. Using set theory: Core only = 1600 (option C).

Source: Adapted from GATE 2024 CS Session 2 Q3

Q5. General Aptitude | 2 marks | 2MCQ

Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix...

Answer: 24

With four distinct elements, the number of possible orders (permutations) is $4! = 24$. Each wizard tries every permutation, so each makes 24 attempts.

Source: Adapted from GATE 2024 CS Session 2 Q2

Q6. General Aptitude | 2 marks | 2MCQ

Consider the relationships among P, Q, R, S, and T: $\ddot{y}y$ P is the brother of Q. $\ddot{y}y$ S is the daughter of Q. $\ddot{y}y$ T is the sister..

Answer: A

P is brother of Q. S is daughter of Q. T is sister of S. R is mother of Q. (1) R is grandmother of S - TRUE. (2) P is uncle of S and T - TRUE.

Source: Adapted from GATE 2025 CS Session 1 Q4

Q7. General Aptitude | 2 marks | 2MCQ

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube...

Answer: B

Folding the given net shows that the faces adjacent in the net become adjacent on the cube exactly as depicted in option B; the opposite faces also match, confirming this configuration.

Source: Adapted from GATE 2025 CS Session 2 Q4

Q8. General Aptitude | 2 marks | 2NAT
Antonym of 'ephemeral':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q10

Q9. Engineering Mathematics | 2 marks | 2MSQ
Probability of at least one head in 3 coin tosses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q11

Q10. Engineering Mathematics | 2 marks | 2MCQ
Derivative of x^3 at $x=2$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q1

Q11. Engineering Mathematics | 2 marks | 2MCQ
Determinant of 2×2 identity:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q1

Q12. Engineering Mathematics | 2 marks | 2NAT
Mean of binomial $n=10, p=0.5$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2NAT
Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q1

Q14. Programming and Data Structures | 1 marks | 1MCQ
Avg hash lookup:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q3

Q15. Programming and Data Structures | 1 marks | 1MCQ
Infix to postfix uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q6

Q16. Programming and Data Structures | 1 marks | 1MCQ
Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q17. Programming and Data Structures | 1 marks | 1MCQ
What is the time complexity of inserting at the beginning of a linked list?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q3

Q18. Algorithms | 1 marks | 1MCQ
BFS uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q7

Q19. Algorithms | 1 marks | 1NAT
Merge sort space:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q7

Q20. Algorithms | 1 marks | 1MCQ
Worst-case time complexity of merge sort:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q12

Q21. Algorithms | 1 marks | 1MCQ
LCS uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 CS Session 1 Q7

Q22. Algorithms | 2 marks | 2NAT
The time complexity of Dijkstra's algorithm using a binary heap is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q7

Q23. Algorithms | 1 marks | 1NAT
Kruskal uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q2

Q24. Algorithms | 1 marks | 1MCQ
Time complexity of binary search:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q11

Q25. Operating Systems | 1 marks | 1NAT
TLB stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 CS Session 1 Q9

Q26. Operating Systems | 2 marks | 2MCQ
*Using 3 frames, which page replacement algorithm would not cause any page fault for reference string
0,1,2,3,0,1,4,0,1,2...*

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q8

Q27. Operating Systems | 1 marks | 1MCQ
Round-robin uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q4

Q28. Operating Systems | 1 marks | 1MCQ
Which of the following is NOT a necessary condition for deadlock?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q4

Q29. Operating Systems | 1 marks | 1MCQ
Inode contains:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q4

Q30. Operating Systems | 1 marks | 1MCQ
Semaphore with value 1 is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q8

Q31. Operating Systems | 1 marks | 1MCQ
Banker's algorithm is for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q4

Q32. Computer Networks | 1 marks | 1NAT
OSPF is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q6

Q33. Computer Networks | 1 marks | 1MCQ
Asymmetric encryption:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q5

Q34. Computer Networks | 1 marks | 1NAT
Which protocol operates at the application layer of the TCP/IP model?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q6

Q35. Computer Networks | 1 marks | 1NAT
CSMA/CD is used in:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q13

Q36. Computer Networks | 1 marks | 1MCQ
TCP connection establishment uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q14

Q37. Computer Networks | 1 marks | 1MCQ
Protocol to find MAC from IP:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q6

Q38. Digital Logic | 1 marks | 1NAT
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q39. Digital Logic | 1 marks | 1MCQ
Full adder has how many inputs:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q2

Q40. Digital Logic | 1 marks | 1MCQ
The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q41. Theory of Computation | 1 marks | 1NAT
Which language is regular?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q13

Q42. Theory of Computation | 1 marks | 1MCQ
Halting problem:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q8

Q43. Theory of Computation | 1 marks | 1MCQ
PDA recognizes:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q14

Q44. Compiler Design | 1 marks | 1MCQ
3-address code operands:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q9

Q45. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q46. CS | 2 marks | 2NAT
Consider routing table of an organization's router shown below: Subnet Number Subnet Mask Next Hop 12.20.164.0 255.255.255.0 12.20.168.0 255.255.255.0 12.20.171.0 255.255.255.0

Answer: 12.20.164.0/22, 12.20.168.0/22

The subnets span from 12.20.164.0 to 12.20.171.255. This range can be exactly covered by two /22 blocks:

12.20.164.0/22 (covers 164 167) and 12.20.168.0/22 (covers 168 171). No larger prefix can cover the range without including addresses outside the given subnets.

Source: Adapted from GATE 2022 CS Session 1 Q55

Q47. CS | 2 marks | 2MCQ
A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stall...

Answer: B

25% load/store. Ideal CPI=2. Memory stall=2 cycles/load-store. Effective CPI = $2 + 0.25 \times 2 = 2.5$.

Source: Adapted from GATE 2024 CS Session 1 Q56

Q48. CS | 2 marks | 2MSQ

Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size),...

Answer: B

TCP: cwnd=12 MSS. After timeout: ssthresh=6, cwnd=1. In slow start, cwnd doubles each RTT.

Source: Adapted from GATE 2024 CS Session 2 Q54

Q49. CS | 2 marks | 2NAT

Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mod...

Answer: System call, hardware interrupt, exception (trap)

A transition from user mode to kernel mode occurs when the CPU must execute privileged operations. This happens on a system call (software request for OS services), on a hardware interrupt (external event requiring OS handling), and on an exception or trap (faults such as page faults or illegal instructions). All three mechanisms guarantee a mode switch to kernel mode.

Source: Adapted from GATE 2023 CS Session 1 Q23

Q50. CS | 2 marks | 2MCQ

In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side...

Answer: D

Square side 2cm. Rectangle PLMN with L on QR. Using geometry/area calculation: area = 4.

Source: Adapted from GATE 2025 CS Session 2 Q9

Q51. CS | 2 marks | 2MCQ

Consider a sequence a of elements $a_0 = 1$, $a_1 = 5$, $a_2 = 7$, $a_3 = 8$, $a_4 = 9$, and $a_5 = 2$. The following operations are perfo...

Answer: B

Stack: push 1,5,7,8,9,2 then pop twice !' 9,8. Queue: enqueue 2,9,8 then dequeue !' 2,9,8. Combined result matches option B.

Source: Adapted from GATE 2023 CS Session 1 Q59

Q52. CS | 2 marks | 2MCQ

Consider the following C program: `#include <stdio.h> void fX(){ void fX(); char a; int main(){ if((a=getchar()) != '\n')...`

Answer: B

Program reads character and recursively calls fX() until newline, then prints the character. Terminates with that single character output.

Source: Adapted from GATE 2024 CS Session 1 Q19

Q53. CS | 2 marks | 2MCQ

Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B...

Answer: A

Two coins: $P(HH)=1/4$, $P(HT)=1/4$, $P(TH)=1/4$, $P(TT)=1/4$. $A=\{HH\}$, $B=\{HH,HT\}$, $C=\{HT,TH\}$. $P(A \cap B)=1/4=P(A)P(B)$. So A and B are independent.

Source: Adapted from GATE 2023 CS Session 1 Q53

Q54. CS | 2 marks | 2MCQ

Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation on the ...

Answer: B

Surjective $f: A \rightarrow B$. Equivalence: $a \sim a'$, iff $f(a \cdot a') = f(a, a')$. The induced function $F: (A/\sim) \rightarrow B$ is onto.

Source: Adapted from GATE 2023 CS Session 1 Q49

Q55. CS | 2 marks | 2MSQ

Consider the following definition of a lexical token id for an identifier in a programming language, using extended re...

Answer: A

Identifier regex: `letter[letter|digit]*`. Check which strings match this pattern. Option A matches.

Source: Adapted from GATE 2023 CS Session 1 Q19

Q56. Digital Logic | 1 marks | 1MCQ

The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q57. CS | 2 marks | 2MCQ

The symbol $\rightarrow$ indicates functional dependency in the context of a relational database. Which of the following options is/are...

Answer: B

Functional dependency $X \twoheadrightarrow Y$: Armstrong's axioms. Given FDs, check which option is TRUE using reflexivity, augmentation, transitivity.

Source: Adapted from GATE 2024 CS Session 1 Q44

Q58. CS | 2 marks | 2NAT

Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i , respectively. Consider ...

Answer: T1 ! T3 ! T4 ! T2

The precedence graph from the given schedule yields edges $T4 \rightarrow T2$, $T3 \rightarrow T2$, $T1 \rightarrow T3$, $T1 \rightarrow T4$, and $T3 \rightarrow T4$. The only topological ordering satisfying all these constraints is $T1$ before $T3$ before $T4$ before $T2$, making the serial schedule $T1 / T3 / T4 / T2$ conflict equivalent to S .

Source: Adapted from GATE 2022 CS Session 1 Q39

Q59. CS | 2 marks | 2MCQ

Consider the following statements: (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the co...

Answer: A

ARP maps IP to MAC (TRUE). RARP maps MAC to IP (TRUE). Both statements are correct.

Source: Adapted from GATE 2025 CS Session 2 Q16

Q60. CS | 2 marks | 2NAT

Consider the following C program: `#include <stdio.h> void fX(){ void fX(); char a; int main(){ if((a=getchar()) != '\n')...`

Answer: 1

Same C program as Q52. For input 'a' followed by newline, number of recursive calls = 1.

Source: Adapted from GATE 2024 CS Session 1 Q19

Q61. CS | 2 marks | 2MCQ

The number of distinct minimum-weight spanning trees of the following graph is _____
b 3 c 1 2 2 1 a d 2 g 2 1 2 1 2...

Answer: 3

By evaluating all possible spanning trees and selecting those with the minimum total weight, we find three distinct trees achieve that minimum weight.

Source: Adapted from GATE 2024 CS Session 2 Q59

Q62. CS | 2 marks | 2NAT

Let $= (, ,)$ be a context-free grammar in Chomsky Normal Form with $= \{ , , \}$ and containing 10 variable symbols including...

Answer: 11

In Chomsky Normal Form, a derivation tree for a string of length n is a full binary tree with n leaves (terminal productions) and $n - 1$ internal nodes (binary productions). Each node corresponds to a production application, so total steps = $(n - 1) + n = 2n - 1$. Here the string 303030 has length 6, giving $2*6 - 1 = 11$ steps.

Source: Adapted from GATE 2024 CS Session 1 Q59

Q63. General Aptitude | 1 marks | 1MCQ
If $f(x) = -f(-x)$ for all real values of x , which one of the following statements is true?

Answer: A

The relation $f(x) = -f(-x)$ holds for all real x , which characterises an odd function. For any odd function, $f(0) = -f(0) \Rightarrow f(0) = 0$. This is the only statement guaranteed to be true, matching option A.

Source: Adapted from GATE 2025 CS Session 2 Q3

Q64. CS | 2 marks | 2MSQ
Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree of G has an even number of odd weight edges.

Answer: D

Let O be the set of odd weight edges. For any spanning tree T , the condition that its total weight is even means $|T \cap O|$ is even. If we take an edge $e \in T$ and add it to T , we obtain a unique cycle $C = T + e$. Replacing any edge f of C by e yields another spanning tree $T' = T - f + e$, which must also have an even number of odd edges. Hence $\text{parity}(e) = \text{parity}(f)$ for every $f \in C$. Therefore all edges of any cycle have the same parity (all even or all odd). This proves statement /D.

Statements /A, /B and /C are not necessary; counter examples exist (e.g., a graph where all edges are odd and the number of vertices is odd satisfies the premise but violates A and B, and a graph consisting of two cycles, one all even and one all odd, satisfies the premise but violates C and B).

Source: Adapted from GATE 2024 CS Session 2 Q51

Q65. CS | 2 marks | 2MSQ
A meld operation on two instances of a data structure combines them into one single instance of the same data structure....

Answer: B

Meldable heaps: Pairing heap and Fibonacci heap support meld in $O(1)$. Binary heap needs $O(n)$. Option B correctly identifies these.

Source: Adapted from GATE 2025 CS Session 2 Q38